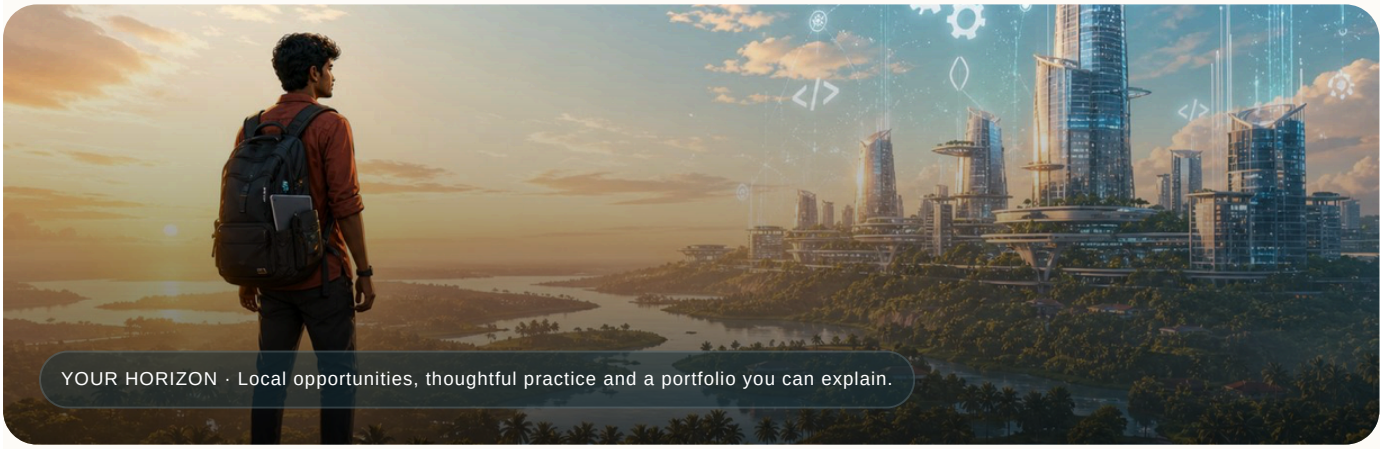
A PRACTICAL DATA ANALYTICS ROADMAP / 2026

Your next chapter. Data Analyst.

Build job-ready analytics skills alongside work, study or other commitments. A flexible roadmap for **South-East London, Brighton and the wider South East.** Excel, SQL and Power BI first. Turn practice into a portfolio you can explain.

Pure data analytics | Shift-friendly plan | UK skills & role guidance ↗

TIMELINE	WEEKLY BUDGET	TOTAL PRACTICE	PORTFOLIO
28 weeks	**12** hours	**336** hours	**05** projects
Flexible—not a hiring deadline	Fits around a changing rota	Learning + building + review	Lead with your best three

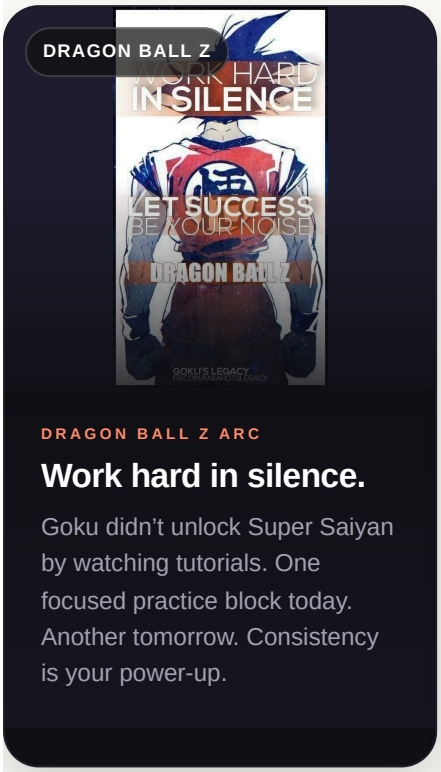


9 training arcs · 142 checkpoints · 5 projects · free core learning resources.

TRAINING ARC

Every session makes you stronger.

Start small. Practise the fundamentals. Keep going when a task takes more than one attempt.



DRAGON BALL Z ARC

Work hard in silence.

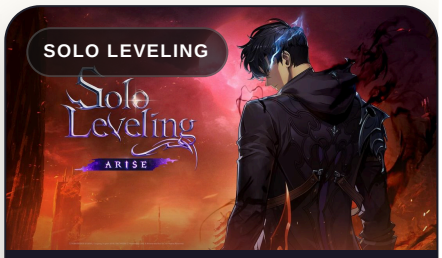
Goku didn’t unlock Super Saiyan by watching tutorials. One focused practice block today. Another tomorrow. Consistency is your power-up.



NARUTO ARC

Zero talent? Perfect.

Rock Lee trained the fundamentals. Start where you are and build through repetition—one checked query, one clearer explanation, one stronger skill at a time.



SOLO LEVELING ARC

E-rank → S-rank.

Day 1: a basic SELECT. Later: a dashboard you can defend. Level up through completed work, not finished playlists.

MARKET INTEL / SOUTH EAST

The right stack. A sharper focus.

Eight advert profiles reviewed: five junior or entry-route examples and three experienced-role comparators. This is a qualitative sample—not a hiring census or a vacancy count.

01 CORE SKILL

Excel + Power Query

EDF's Hove reporting profile and Kreston Reeves' junior specification emphasise spreadsheet, quality and communication skills.

View evidence ↗

02 CORE + DIALECT BRIDGE

SQL + a T-SQL bridge

Barts' junior profile names MS SQL Server; the local Oxleas experienced-role profile also uses the Microsoft reporting stack. Learn portable SQL, then bridge the dialect.

View evidence ↗

03 CORE + REPORT OPERATIONS

Power BI + operations

A useful report needs more than charts. Learn refresh, workspace access and row-level security alongside data preparation, modelling and DAX.

View evidence ↗

04 ROLE-DEPENDENT

Python + pandas

The indexed 7IM junior specification calls Python highly beneficial. EDF and Kreston examples do not establish it as a universal entry prerequisite.

View evidence ↗

05 CORE, NOT OPTIONAL

Quality + communication

Kreston's junior role explicitly includes data audits, GDPR, documentation and clear reporting. Evidence of trustworthy work matters alongside tools.

View evidence ↗

06 TARGETED, NOT AUTOMATIC

PL-300 + extra tools

7IM's historical junior profile expected PL-300 during probation. That is an employer-specific signal—not proof that every applicant must buy a certificate.

View evidence ↗

Salary context, not a promise: Prospects gives typical UK starting salaries of £28,000–£32,000. Local jobs vary; London-weighted NHS bands and experienced-role pay are not interchangeable with junior starting salaries. Prospects ↗

Open your reviewed tool-priority table

SKILL	PRIORITY	WHEN	DEPTH FOR YOU
Excel / Power Query	CORE	Weeks 2–4	Refreshable tables, formulas, pivots, combine files, exception checks.

SKILL	PRIORITY	WHEN	DEPTH FOR YOU
SQL	CORE	Weeks 5–8	Joins, CTEs, NULLs, windows, dates, grain and validation.
SQL Server / T-SQL	BRIDGE	Inside weeks 5–8	Translate a few queries; Microsoft Learn supplement. Not another full course.
Power BI / DAX	CORE	Weeks 9–13	Model, measures, reporting, RLS, refresh, access and accessibility.
Data quality / privacy / communication	CORE	Throughout	Metric definitions, checks, UK GDPR awareness and usable explanations.
GitHub / documentation	SUPPORT	Week 1 onward	README, versions and safe publication; no advanced Git requirement.
Python / pandas	SHOULD	Weeks 16–19	Useful for automation and some London roles; not every entry post.
NHS definitions / reporting	TARGETED	Within project time	Only when applying to healthcare roles with those essential criteria.
PL-300	OPTIONAL	After projects / if target asks	Current official exam guide, not just a course certificate.
Power Automate / VBA / SSRS	OPTIONAL	After core; advert-led	Awareness of legacy/automation stack; do not add three full courses.
Tableau / GA4 / cloud / dbt / ML	DEFER	Role-specific later	Consider only for a different specialism or repeated essential criteria.

What employers look for.

Explore example roles, the skills they ask for and the experience needed. Check each advert's status before applying.

SKILLS THAT MATTER

Focus on useful, demonstrable skills.

Clean data, reliable queries, clear reports and sound judgement. Build evidence that shows how you work—not just a list of courses you have watched.

08

advert profiles reviewed, not vacancies

01 **EDF**

Entry-route example

Reporting Analyst

Brighton / Hove

Level: Entry-friendly reporting example; not explicitly titled junior

Status: Historical: advertised closing date 2 March 2026; employer page access limited

Evidence access: Employer search-index excerpt; direct page access limited

What the profile says: Excel formulas and data manipulation; Power BI learning; accuracy, dashboards and stakeholder communication

What this means for your study: Supports an Excel/reporting-first application route. No evidence that every entry role requires Python or SQL.

Employer / NHS source ↗

02 **Legal & General**

Entry-route example

Data Analyst Apprenticeship 2026

Hove

Level: Apprenticeship entry route

Status: Expired in indexed employer page; September 2026 intake. Employer page access limited

Evidence access: Detailed employer search-index text; direct page access limited

What the profile says: Analysis, coding, visualisation, automation and non-technical communication; qualification, residency and right-to-work conditions

What this means for your study: Add apprenticeships as an optional entry route. Check each future advert and current funding rules; do not generalise this advert's residency condition to every apprenticeship.

Employer / NHS source ↗

03

Kreston Reeves

Entry-route example

Junior Data Quality & Reporting Analyst

Canterbury, Kent — adjacent-market comparator, not Brighton or SE London

Level: Junior, 12-month fixed term

Status: Employer page accessible with Apply Now on review date; recheck before applying

Evidence access: Full employer advert read

What the profile says: Advanced Excel, PivotTables, Power Query, formulas, cleansing, audit trails, GDPR, documentation and stakeholders; no explicit SQL/Python requirement in the read specification

What this means for your study: Strengthen data-quality evidence, documented checks and privacy. Canterbury is a separate commute decision; no salary inferred from aggregator estimates.

Employer / NHS source ↗

04

Barts Health NHS Trust

Entry-route example

Junior Information Analyst Returns

Canary Wharf E14 — East London, not South-East London

Level: Band 5 junior

Status: NHS Jobs explicitly says closed; posted 29 September 2026

Evidence access: Full NHS Jobs advert and displayed person specification read

What the profile says: MS SQL Server and BI tools, reporting, quality and communication; NHS standard definitions and reporting knowledge essential; automation tools desirable; advert also lists Visual Basic/HTML among technology experience

What this means for your study: Add a short SQL Server/T-SQL bridge and an optional NHS domain pack. A general portfolio does not automatically satisfy NHS essential criteria; do not turn one advert's legacy technologies into a universal syllabus.

Employer / NHS source ↗

05

Seven Investment Management

Entry-route example

Junior Data Analyst

London / City — wider-London comparator, not verified SE London base

Level: Junior

Status: Employer URL now redirects to Vacancy not found; historical indexed specification only

Evidence access: Detailed employer search-index text; employer page no longer available

What the profile says: Power BI, Excel, numerical and concise communication skills; Python highly beneficial; expected PL-300 within six-month probation; basic financial-services context

What this means for your study: Keep Python as a useful differentiator. PL-300 can be employer-specific after hiring, not a universal prerequisite or a promised sponsored exam.

Employer / NHS source ↗

06

Oxleas NHS Foundation Trust

Senior comparator

Performance and Information Analyst

Dartford DA2; trust serves Bexley, Bromley and Greenwich

Level: Experienced Band 7 — not junior

Status: NHS Jobs explicitly closed; posted 4 March 2026

Evidence access: Full advert duties and experience requirement read

What the profile says: Minimum three years SQL/database/warehouse experience; SQL, SSRS, Excel, Power BI; data quality and staff supervision

What this means for your study: Local ecosystem signal only. Do not require three years' experience, stored procedures/triggers or people management before beginning junior applications.

Employer / NHS source ↗

07

Lewisham and Greenwich NHS Trust

Senior comparator

Advanced Information Analyst

Catford SE6 — South-East London

Level: Advanced Band 7 — not junior

Status: Advert closing date 17 May 2026; historical

Evidence access: Detailed NHS Jobs search result / person-specification text read

What the profile says: Advanced SQL and Excel, Python/R forecasting, Power BI/SSRS, NHS datasets and information governance; significant experience

What this means for your study: Confirms relevant local stack but advanced forecasting, optimisation and expert SQL are later specialisms, not a junior baseline.

Employer / NHS source ↗

08

MetLife

Senior comparator

Senior Data and Insights Analyst (body: Senior Data and Reporting Analyst)

Brighton

Level: Senior — preferred 3–5+ years

Status: Employer page accessible; posted 2 October 2026; not counted as a junior opportunity

Evidence access: Full employer responsibilities and requirements read

What the profile says: Excel, Power Query, Power BI/DAX, SQL, governance, lineage, controlled change and automation; Python/Power Automate preferred; legacy VBA advantageous

What this means for your study: Add lightweight report handover/change-log concepts now. Keep Power Automate, VBA and advanced platform engineering optional, not compulsory junior tools.

Employer / NHS source ↗

Limits of the evidence: no claim of representative skill percentages, salary premiums or interview conversion rates. Most examples are historical or closed. SE London’s two detailed local profiles here are experienced roles; they are stack signals, not proof of a large junior market. Canary Wharf is East London and Canterbury is Kent.

LEARNING RESOURCES

Learn with purpose.

Choose one main course per tool. Use focused lessons to fill gaps, then practise without the tutorial.

Study rule: one primary course per tool, then targeted gap lessons. The video budget is inside the 336 hours—not added on top. Paid problem packs, course certificates and subscriptions are not required to follow this roadmap.

01

EXCEL · CORE

Excel for Data Analytics

Primary learning link ↗ Outline / files ↗

▼ **Coverage, gaps and exactly what to study**

WHAT TO EXPECT

Formulas/functions, charts, tables, PivotTables, Power Query, Power Pivot and two projects are present in the published outline.

YOUR ESSENTIAL STUDY ROUTE

Prioritise chapters 2–5 and 7. Add UK date parsing, reconciliation, exception logs and a repeatable month-file refresh. Revisit basics only if your diagnostic fails.

DEFER / AVOID DUPLICATION

Solver, Scenario Manager, Goal Seek and deeper Power Pivot are useful later, not blockers before junior reporting applications.

ACCESS AND ACCURACY NOTES

The creator lists free videos/files but paid supporter problems and certificates. If individual Excel files fail, download the repository ZIP or use the creator’s alternate file link.

02

FOUNDATIONS + FOCUSED PRACTICE

SQL foundations + selected intermediate

Primary learning link ↗ Outline / files ↗

▼ **Coverage, gaps and exactly what to study**

WHAT TO EXPECT

Start with analyst-focused SQL foundations: filtering, joins, aggregations, CASE and CTEs. Use selected intermediate lessons for window functions, dates and data cleaning, then practise translating queries to T-SQL.

YOUR ESSENTIAL STUDY ROUTE

Use Luke's foundations course for analyst queries, CASE and CTEs. Supplement window/date/cleaning topics with selected Intermediate SQL lessons ↗ and the official T-SQL window lesson.

DEFER / AVOID DUPLICATION

Do not spend time on triggers, database administration or a second full beginner course. Keep one primary database and a short dialect bridge.

ACCESS AND ACCURACY NOTES

PostgreSQL is the main course dialect. The free starter job dataset is historical 2023 data, not evidence of today's London or Brighton market. Supporter exercises/certificates are paid.

03

POWER BI · CORE

Power BI for Data Analytics

Primary learning link ↗ Outline / files ↗

▼ **Coverage, gaps and exactly what to study**

WHAT TO EXPECT

Published modules include Grand Tour, visualisations, Power Query and DAX, plus star-schema/dashboard projects.

YOUR ESSENTIAL STUDY ROUTE

Explicitly add workspace roles/access, gateway and scheduled refresh concepts, a Desktop RLS test, Performance Analyzer, mobile layout and accessibility. Use current Microsoft Learn guidance.

DEFER / AVOID DUPLICATION

Do not assume the creator's "PL-300 aligned" description guarantees full current exam coverage. Newer exam topics need a separate exam checklist if you choose certification.

ACCESS AND ACCURACY NOTES

Use the creator's complete course-file link ↗; the repository warns that some large files are missing. Some Service exercises require a suitable account/licence; no paid sharing is required for the portfolio.

04

PYTHON · AFTER THE CORE

Python for Data Analytics

Primary learning link ↗ Outline / files ↗

▼ **Coverage, gaps and exactly what to study**

WHAT TO EXPECT

The outline covers basics, pandas, NumPy, Matplotlib, advanced cleaning/manipulation and a project.

YOUR ESSENTIAL STUDY ROUTE

Prioritise functions, dates, CSVs, missing values, groupby, merge validation, readable plots and reproducible export. Add one small test fixture and rerun from a clean environment.

DEFER / AVOID DUPLICATION

Defer deep OOP, machine learning, scraping and advanced plotting unless target roles justify them. Do not postpone Excel/BI applications until every Python chapter is complete.

ACCESS AND ACCURACY NOTES

Use current notebooks when package APIs differ from the video. Free videos/code do not imply every practice problem or certificate is free.

05

STATISTICS · APPLIED

Statistics + official guidance

Primary learning link ↗ Outline / files ↗

▼ **Coverage, gaps and exactly what to study**

WHAT TO EXPECT

Khan Academy is a broad reference, not a role-specific analyst syllabus; use only the topics your checkpoint requires.

YOUR ESSENTIAL STUDY ROUTE

Practise denominators, weighted averages, percent vs percentage points, bias, uncertainty, correlation and plain-English communication on your own data.

DEFER / AVOID DUPLICATION

Do not add a full probability or data-science qualification. Introductory test reasoning is enough unless a role asks for deeper inference.

ACCESS AND ACCURACY NOTES

Complement learning with ICO principles ↗ and the published methodology for the public dataset you use.

If you choose PL-300: the separate exam boundary

Microsoft's skills guide effective 20 April 2026 covers Prepare data (25–30%), Model data (25–30%), Visualize/analyze (25–30%), and Manage/secure (15–20%). The general roadmap is **not a complete current exam-prep course**.

Before booking, check every current objective, including the newer Import/DirectQuery/DirectLake distinctions, calculation groups, visual calculations, Copilot features, workspace governance and sensitivity/access controls. Some features require suitable accounts or capacity. Do not buy platform capacity just to keep up with an optional study plan.

Current official PL-300 skills guide ↗

SYLLABUS AT A GLANCE

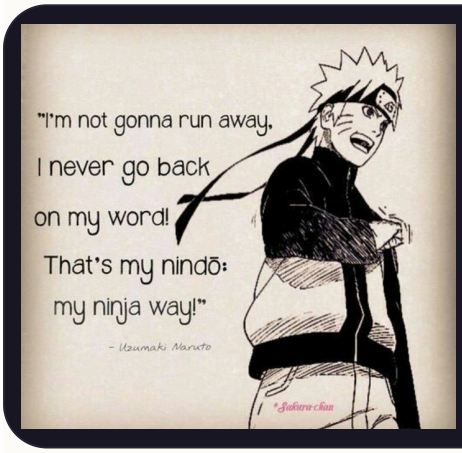
The skills to practise—and how to prove them.

Use this checklist to connect each skill with a practical piece of work. Keep role-specific additions separate from the core path.

MODULE / SKILL	STUDY FOCUS	WHERE	EVIDENCE TO PRODUCE
Excel formulas, lookups, pivots	Core practice	Excel	Refreshable workbook, checked against raw totals.
Excel exceptions / combine files	Practical application	Excel	Exceptions tab + source-file field + tested refresh.
SQL joins, NULLs, dates, CASE	Core practice	SQL	Explain grain and reconcile joins.
CTEs and window functions	Foundations + targeted lessons	SQL	Selected intermediate lessons + 30-query project.
SQL Server / T-SQL	Short dialect bridge	SQL	Translate five queries; not another full database course.
Star schema / DAX / date table	Core practice	Power BI	Validated model and five reconciled measures.
Refresh / gateway / workspace access	Hands-on practice	Power BI	Explain setup and failure modes; use mock runbook if no Service account.
Row-level security / safe sharing	Hands-on practice	Power BI	Desktop View as test and access explanation.
Performance / accessibility / mobile	Hands-on practice	Power BI	One measured visual and a tested mobile layout.
Stats / denominators / uncertainty	Apply to real questions	Statistics	One-page briefing with a limitation.
UK GDPR / responsible AI	Hands-on practice	Statistics	Document permitted data use and independent validation.
Stakeholder question / acceptance checks	Hands-on practice	Statistics / capstone	Two acceptance criteria and an external check; analytics practice, not BA training.
pandas / automation / tests	Reproducible practice	Python	Known-output fixture + reproducible script.
Lineage / handover / change log	Hands-on practice	Capstone	Owner, source, metric definitions, refresh, checks and version history.
Person-spec evidence / supporting statement	Hands-on practice	Interview / job hunt	Map claims to criteria and record genuine gaps.
NHS domain definitions	Role-specific study	Job hunt / project	Read the NHS Data Dictionary and dataset methodology for the role.

MODULE / SKILL	STUDY FOCUS	WHERE	EVIDENCE TO PRODUCE
Finance / retail context	Domain practice	Portfolio	Metric glossary; label assumptions and synthetic data.
Power Automate / VBA / SSRS	Optional awareness	After core	Investigate only when repeated target adverts justify it.
ML / full-stack / cloud engineering	Outside the core path	Later specialisms	Not required by this reporting-first plan.

Use the syllabus alongside job criteria: this roadmap covers core junior reporting and analytics skills. Employers may also require domain knowledge, specific systems or prior experience. Check the essential criteria for each role. Certification preparation has its own syllabus.



THE NINDŌ RULE

“I never go back on my word. That’s my ninja way.”

Use a rule, not a mood: avoid two unplanned skips in a row. After an exhausting shift, make the next session a ten-minute review. Planned rest and illness are not failures.

THE TRAINING ARC

Your 28-week roadmap.

Practise each skill, then complete the boss task. Select the course chapters you need and spend the rest of the session building, checking and explaining your work.

Academy Arc · WEEK 1

01 Setup, workflow & proof of work

0/12 · 0%

12h total learning + building

Check your starting point, set up your tools and choose a regular study routine. Use the diagnostic to identify what you already know and what needs practice.

Boss to defeat: A working toolkit, a public learning-log repository and one short analysis of a clean sample CSV.

WATCH / READ · ONE MAIN RESOURCE, THEN PRACTISE

GitHub Hello World ↗
Official guide · free

Get Power BI Desktop ↗
Official Windows requirements & installation

- [ ] Check your device: Power BI Desktop requires Windows; arrange Windows access if using a Mac.
- [ ] Set up Excel or Microsoft 365 access; identify which Power Query features your version supports.
- [ ] Choose one SQL learning environment: browser for basics, PostgreSQL + DBeaver/VS Code for the main course. Use SQL Server only for the short Microsoft bridge.
- [ ] Create folders: raw-data, cleaned-data, sql, notebooks, reports and screenshots.
- [ ] Create GitHub profile + learning-log repository; upload a README and a sample SQL file.
- [ ] Learn Git status, add, commit, pull and push, or use GitHub's web upload first.
- [ ] Set a repeatable 12-hour weekly schedule around your work, study and other commitments.
- [ ] Record your baseline: one PivotTable, one SELECT query and a 60-second explanation.
- [ ] Use only public or synthetic practice data; never upload confidential employer, customer or colleague information.
- [ ] Write one analysis question with a metric, time period and intended reader.
- [ ] Identify personal/sensitive data, check the permitted purpose, and use only approved public or synthetic practice files.
- [ ] Record ten genuine employer adverts: verify location, level, essential criteria, closing date and whether it is a paid job rather than a fee-charging course.

Training rule: If setup takes more than a session, use browser tools and keep practising. Tool installation is not the boss.

03 h learn **07** h build / practise **02** h check / review

Foundation Arc · WEEK 2–4

02 Excel Pro & Power Query

0/16 · 0%

36h total learning + building

Build on what you already know. Spend less time watching familiar material and more time explaining how you checked the result.

Boss to defeat: Project 1: a refreshable retail workbook with raw/clean tabs, PivotTables and a one-page recommendation.

WATCH / READ · ONE MAIN RESOURCE, THEN PRACTISE

Excel for Data Analytics — Full Course ↗
Luke Barousse · ~11h · English · watch selected chapters

Excel help & learning ↗
Microsoft · reference

- [ ] Clean like a pro: Duplicates, TRIM, flash fill
- [ ] Formulas: IF, SUMIFS, TEXT/DATE
- [ ] Lookups: VLOOKUP → XLOOKUP → INDEX/MATCH
- [ ] PivotTables: Group by month, slicers, PivotChart
- [ ] Power Query: Unpivot, merge, refresh
- [ ] Dashboard: 1 interactive dashboard answering a business Q
- [ ] Use Excel Tables, structured references and consistent data types.
- [ ] Handle blanks, errors and dates with UK locale; distinguish missing values from zero.
- [ ] Check XLOOKUP exact matches and unmatched keys; explain one-to-many lookup limits.
- [ ] Calculate revenue, units, average basket and percentage change with explicit definitions.
- [ ] Reconcile row counts and sales totals before and after cleaning.
- [ ] Write three findings and one limitation; label synthetic data clearly.
- [ ] Close the tutorial and rebuild your main PivotTable without help.
- [ ] Export a readable dashboard PDF and document refresh steps.
- [ ] Build an exceptions tab showing failed checks; add conditional formatting and data validation where useful.
- [ ] Combine two monthly CSV files with Power Query, keep a source-file column, and test a later refresh.

Training rule: Your boss test: someone else can replace the CSV and refresh the report without calling you.

08 h learn **24** h build / practise **04** h check / review

Chakra Control Arc · WEEK 5–8

03 SQL for reliable answers

0/19 · 0%

48h total learning + building

Use SQL to retrieve, combine and validate data. Practise the reasoning behind the query, not just the syntax.

Boss to defeat: Project 2: a small retail database, 30 documented queries and a 30-minute closed-book SQL test.

WATCH / READ · ONE MAIN RESOURCE, THEN PRACTISE

SQL for Data Analytics — foundations + project ↗
Luke Barousse · ~4h · PostgreSQL; free video and starter files

Intermediate SQL — selected window/date/cleaning lessons ↗
Luke Barousse · ~6h full video; select gaps only, not a second compulsory binge

Window functions in T-SQL ↗
Microsoft Learn · free official gap lesson

Short SQL Server / T-SQL bridge ↗
Microsoft Learn · select relevant querying modules; optional local setup

- [ ] Foundations: SELECT, WHERE, ORDER BY, NULLs
- [ ] Aggregations: COUNT/SUM/AVG, GROUP BY, HAVING, CASE
- [ ] JOINs: INNER/LEFT/FULL/UNION — understand preserved rows
- [ ] Window: ROW_NUMBER, RANK, LAG/LEAD, PARTITION BY
- [ ] Project: 10 queries to GitHub (retail or Northwind sample)
- [ ] Understand primary/foreign keys, row grain and duplicate keys.
- [ ] Filter dates safely with half-open intervals and explicit date formats.
- [ ] Use IS NULL, COALESCE and NULLIF; explain COUNT(*) vs COUNT(column).
- [ ] Demonstrate how a one-to-many join can inflate totals.
- [ ] Write and explain a subquery and a CTE solving the same problem.
- [ ] Find top products per category with a ranking window.
- [ ] Calculate month-on-month change with LAG and handle missing months.
- [ ] Find unmatched rows with LEFT JOIN + IS NULL or NOT EXISTS.
- [ ] Explain WHERE vs HAVING; UNION vs UNION ALL.
- [ ] Save 30 queries with questions, expected outputs and reconciliation checks.
- [ ] Complete a timed test without copying; review mistakes the following day.
- [ ] Translate five existing queries to T-SQL and explain TOP vs LIMIT, date functions and FULL OUTER JOIN support.
- [ ] Calculate a running total with an explicit ROWS frame and explain how ties affect ranking.
- [ ] Read a simple query plan and distinguish a view from a temporary table; leave production tuning and database administration for later.

Training rule: Use PostgreSQL for the main course. Spend a short bridge session translating TOP/LIMIT, GETDATE/CURRENT_DATE and DATEADD/DATEDIFF; do not install or master multiple database servers at once.

10 h learn **32** h build / practise **06** h check / review



GO BEYOND · PLUS ULTRA

“Go beyond. PLUS ULTRA!”

You can now retrieve and check data. Next: turn it into a dashboard that answers a real question. One validated measure is a bigger power-up than ten decorative visuals.

Power-up Arc · WEEK 9–13

04 Power BI, modelling & DAX

0/22 · 0%

60h total learning + building

The hero skill is not adding charts. It is producing trustworthy measures and a page that helps a manager make one decision.

Boss to defeat: Project 3: a retail performance dashboard with a star schema, validated measures and a 3-minute demo.

WATCH / READ · ONE MAIN RESOURCE, THEN PRACTISE

Power BI for Data Analytics — Full Course ↗
Luke Barousse · ~8h25 · English

Star schema guidance ↗
Microsoft · official

DAX reference ↗
Microsoft · official

Service, security & refresh gap pack ↗
Microsoft Learn · current skills guide, not a claim of full exam readiness

- [ ] Get Data: Power Query clean/unpivot
- [ ] Model: Star schema, date table
- [ ] DAX: SUM, DIVIDE, CALCULATE and filter context
- [ ] Visuals: KPI cards, slicers, drill-through and accessible labels
- [ ] Publish: Understand Service, sharing and refresh limitations
- [ ] Portfolio: Retail dashboard + README + screenshots
- [ ] Build fact sales and product/date/store dimensions with unique keys.
- [ ] Use one-to-many relationships and explain filter direction.
- [ ] Distinguish measures, calculated columns and Power Query steps.
- [ ] Create a complete date table and consistent time intelligence measures.
- [ ] Validate at least five KPI totals against Excel or SQL.
- [ ] Show prior-period comparison only when date coverage is comparable.
- [ ] Add clear units, source date, metric definitions and caveats.
- [ ] Test slicers, drill-through, blank values and zero denominators.
- [ ] Understand that Publish to web is public: only use non-sensitive public/synthetic data.
- [ ] Provide PDF/screenshots + demo if you cannot share a live report.
- [ ] Start a small number of targeted applications once Projects 1–3 are explainable.
- [ ] Explain Import vs DirectQuery, when an on-premises gateway is needed, and what can cause scheduled refresh to fail.
- [ ] Create a simple row-level security role in Desktop and use View as to test it with synthetic data.
- [ ] Explain workspace roles, Viewer vs editing permissions, and why Publish to web is not secure sharing.

- [ ] Use Performance Analyzer once; remove unnecessary columns and investigate a slow visual.
- [ ] Configure a tooltip/bookmark or navigation button, then test keyboard access, contrast and a mobile report layout.

Training rule: Power BI Desktop authoring is free; some sharing and refresh options need licences. A paid subscription is not a prerequisite for portfolio evidence.

12 h learn **40** h build / practise **08** h check / review

Analyst Thinking Arc · WEEK 14–15

05 Statistics, quality & storytelling

0/15 · 0%

24h total learning + building

Use these ideas throughout every project; these two weeks consolidate them. This is analytics, not a machine-learning detour.

Boss to defeat: A one-page decision brief: one question, three checked findings, a recommendation and limitations.

WATCH / READ · ONE MAIN RESOURCE, THEN PRACTISE

Statistics & probability ↗
Khan Academy · free · selected topics

UK official statistics ↗
ONS · metadata and methodology practice

- [ ] Statistics: Mean, median, standard deviation and outliers
- [ ] Correlation: Correlation ≠ causation; confounding
- [ ] Story: Question → evidence → recommendation → limitation
- [ ] Choose an appropriate denominator and avoid averaging averages.
- [ ] Distinguish percentage change from percentage-point change.
- [ ] Recognise sampling bias, selection bias and small sample problems.
- [ ] Explain confidence intervals and uncertainty in plain English.
- [ ] Describe a basic A/B test, control group and random assignment.
- [ ] Check whether seasonality, missing periods or population changes affect comparisons.
- [ ] Use bar/line charts appropriately and avoid misleading axis scales.
- [ ] Define a KPI with unit, grain, owner, period and source.
- [ ] Practise a 90-second explanation for a non-technical manager.
- [ ] Apply a practical UK GDPR checklist: purpose, minimisation, accuracy, retention, access and accountability; escalate uncertainty to the data owner.
- [ ] Define a reporting request with audience, KPI formula, source, deadline and two acceptance checks; this is analyst practice, not a Business Analyst syllabus.
- [ ] If using AI to explain a query, use synthetic examples, verify the output independently and never paste confidential work data into an unapproved tool.

Training rule: Never tick a finding until you can say what would make it wrong.

05 h learn **15** h build / practise **04** h check / review

Automation Arc · WEEK 16–19

06 Python, pandas & reproducibility

0/16 · 0%

48h total learning + building

Add Python after Excel, SQL and Power BI. Do not delay suitable applications because you have not finished a Python playlist.

Boss to defeat: Project 4: a reproducible public-data analysis and a script that turns a raw CSV into a checked summary report.

WATCH / READ · ONE MAIN RESOURCE, THEN PRACTISE

Python for Data Analytics — Full Course ↗
Luke Barousse · ~11h · English · select relevant chapters

pandas introductory tutorials ↗
Official · free

Python tutorial ↗
Official · reference

- [ ] Python: Variables, lists, dictionaries, loops and functions
- [ ] pandas: Read CSV, inspect, filter and select
- [ ] Clean: Missing values, duplicates and data types
- [ ] Analyse: groupby, merge, pivot_table
- [ ] Visualise: Matplotlib or Seaborn with readable labels
- [ ] Project: Notebook + README + reproducible outputs
- [ ] Create a virtual environment and record package requirements.
- [ ] Parse dates explicitly; inspect UK/US date ambiguity.
- [ ] Validate merge keys and check row counts before/after joins.
- [ ] Write reusable functions for load, clean, validate and export.
- [ ] Use assertions for totals, unique keys and non-negative values where appropriate.
- [ ] Export clean CSV + an Excel summary without overwriting raw files.
- [ ] Re-run from a clean environment or a fresh Colab session.
- [ ] Keep secrets and personal data out of notebooks and version control.
- [ ] Create a tiny test fixture with known outputs to check one cleaning function and one aggregation.
- [ ] Document relative paths, package versions and input/output filenames so another person can rerun the script.

Training rule: One working automation is worth more than ten copied notebooks. Measure runtime or manual steps honestly; do not invent hours saved.

10 h learn **32** h build / practise **06** h check / review

Final Boss Arc · WEEK 20–23

07 An end-to-end analytics capstone

0/15 · 0%

48h total learning + building

Combine skills you already learned. Choose a question you can defend, rather than adding more tools.

Boss to defeat: Project 5: an East Sussex public-data brief connecting SQL, a checked dataset and a Power BI dashboard, with a 5-minute walkthrough.

WATCH / READ · ONE MAIN RESOURCE, THEN PRACTISE

ONS datasets ↗
Official public data · select a manageable CSV

East Sussex in Figures ↗
Local statistics & metadata

Find open data ↗
UK data catalogue

- [ ] Choose one question: how has a selected local indicator changed over time?
- [ ] Record source URL, download date, licence, geography and definitions.
- [ ] Write a data dictionary and identify the grain of every table.
- [ ] Keep immutable raw files and scripted cleaning steps.
- [ ] Use SQL to answer at least eight meaningful questions.
- [ ] Cross-check dashboard totals against SQL and an independent Excel calculation.
- [ ] Explain missing data, boundary changes and incomparable time periods.
- [ ] Create a concise dashboard with context, clear units and accessible colours.
- [ ] Write a one-page insight brief, separating evidence from assumptions.
- [ ] Document setup, refresh steps and known limitations in the README.
- [ ] Record a 3–5 minute walkthrough: question, data, checks, insight, action.
- [ ] Ask someone to reproduce one number and incorporate their feedback.
- [ ] Polish your best three projects; keep two smaller ones as supporting evidence.
- [ ] Create a one-page report runbook: owner, metric definitions, lineage, refresh timing, checks and what to do when the source changes.
- [ ] Keep a short change log and have another person test the two acceptance checks before you call the report complete.

Training rule: A reproducible analysis with caveats beats an attractive dashboard with unexplained numbers.

04 h learn **38** h build / practise **06** h check / review

Boss Dialogue Arc · WEEK 24–25

08 SQL tests & interview confidence

0/12 · 0%

24h total learning + building

Build this habit earlier with short weekly practice. These weeks focus on presenting evidence clearly under time pressure.

Boss to defeat: Two timed SQL mocks, one Excel task, one dashboard critique and four truthful STAR stories.

WATCH / READ · ONE MAIN RESOURCE, THEN PRACTISE

SQL interview questions ↗
Practice · some questions may require an account

HackerRank SQL ↗
Practice · free account

- [ ] Explain a LEFT JOIN and demonstrate duplicate-row inflation aloud.
- [ ] Complete two 30-minute SQL mocks and revisit every error.
- [ ] Complete an Excel clean → lookup → pivot task from a fresh file.
- [ ] Explain row context, filter context and CALCULATE using your report.
- [ ] Present your flagship project in five minutes without reading notes.
- [ ] Prepare a truthful 60-second introduction connecting your background, analytics practice and target role.
- [ ] Write four STAR stories: quality, teamwork, prioritisation and difficult customer.
- [ ] State limitations and what you would investigate next.
- [ ] Practise asking clarifying questions before starting a take-home task.
- [ ] Prepare questions on data quality, onboarding, tools and success measures.
- [ ] Rehearse a salary conversation using the advert's range and role scope.
- [ ] Use one person specification to build an evidence-to-criteria matrix, including genuine gaps, before writing a supporting statement.

Training rule: A gap is not a failure: explain what you know, what you would check and how you would test it.

03 h learn **17** h build / practise **04** h check / review

Endgame Arc · WEEK 26–28

09 Applications, referrals & feedback

0/15 · 0%

36h total learning + building

Applications can begin in week 13. Weeks 26–28 are a focused search sprint, not a promise that an offer arrives on a deadline.

Boss to defeat: A sustainable weekly pipeline, tailored CV and project links; iterate on real feedback rather than waiting for perfection.

WATCH / READ · ONE MAIN RESOURCE, THEN PRACTISE

NHS Jobs ↗
Search analyst, information analyst, reporting

Civil Service Jobs ↗
Check essential criteria and application format

Reed: Eastbourne search ↗
Use wider geography and related role titles too

- [ ] Set alerts for Junior Data Analyst, MI Analyst, Reporting Analyst and BI Assistant.
- [ ] Search Eastbourne, Lewes, Brighton/Hove and hybrid roles with a realistic commute.
- [ ] Make a shortlist of ten employers; confirm vacancies on their own careers pages.
- [ ] Write an ATS-readable CV with Excel, SQL, Power BI and project links near the top.
- [ ] Describe your employment status accurately; list a role as current only while you still hold it.
- [ ] Apply to around five well-matched roles per week when available; adjust to supply and rota.
- [ ] Send two personalised networking/referral messages per week without spamming.
- [ ] Track date, criteria, CV version, follow-up, outcome and feedback.
- [ ] Follow up once when appropriate and respect no-contact instructions.
- [ ] After 15–20 matched applications without screens, review CV, eligibility and targeting.
- [ ] Practise weekly while applying; do not restart a new bootcamp after each rejection.
- [ ] Check salary, hours, location, employment status and any right-to-work/sponsorship conditions.
- [ ] Keep learning and searching beyond week 28 if needed; extend the timeline, not the guilt.
- [ ] Create separate saved searches for Brighton/Hove, SE London boroughs, and wider-London roles; assess each commute from your actual home.
- [ ] Check Level 3 Data Technician / Level 4 Data Analyst apprenticeships where eligible; verify each advert’s qualifications, residency/funding and existing-qualification rules.

Training rule: No guaranteed conversion rate. Your targets are controllable actions; interview and offer outcomes are not.

02 h learn **28** h applications / preparation **06** h check / review

REAL-LIFE TRAINING

A routine that fits real life.

Keep the commitment flexible enough to repeat. The 12-hour budget includes projects, revision and job-search work—not extra homework on top.

Standard week · 12 hours

- **Four 90-minute blocks:** 6 hours of focused practice across shift days.
- **One 4-hour build block:** on your most suitable day off; include breaks.
- **Two 1-hour review blocks:** redo mistakes, explain a result, update your log.
- **One genuinely free day:** move blocks to protect rest.

Choose times after seeing your rota. A fixed 05:30 wake-up is not required.

Inside a 90-minute block

- 10 min: retrieve yesterday's idea without notes.
- 20 min: one targeted lesson or documentation page.
- 50 min: build or solve independently.
- 10 min: check your output and record a mistake or insight.

Across the week, aim for roughly 70% doing/checking and 30% learning. This is a study rule, not a scientific guarantee.

Heavy rota / low-energy week: use three 20-minute review sessions and one 60-minute build block (2 hours total). Do not pretend that equals 12 hours. Roll unfinished work forward and revise your finish date. On breaks, rest first; study only if it helps you.

The five laws of your training arc

1. Build from memory after the lesson.
2. One boss per season; fix the deliverable before adding tools.
3. Log evidence, not just hours.
4. Review mistakes in short repeatable sessions.
5. Protect sleep, planned rest and confidential data.

MILESTONE SAVE-POINTS

Look back. You are levelling up.

Advance when the evidence is ready; the week number is a planning guide.

Week 1	Toolkit works, baseline tested, learning log started.
Week 4	Excel retail workbook refreshes and totals reconcile.
Week 8	30 SQL queries explained and one timed test completed.
Week 13	Power BI retail report + demo; first matched applications if ready.
Week 15	One-page decision brief with uncertainty and limitations.
Week 19	Reproducible pandas analysis + a report-generation script.
Week 23	Flagship local-data capstone; best three projects polished.
Week 25	Two SQL mocks, an Excel task and four STAR stories.
Week 28	A functioning application pipeline and next-step review—not a guaranteed offer.



BOSS RUSH · SOLO LEVELING

ARISE. Five projects. Five levels.

Clear one, unlock the next. Three polished case studies will lead your portfolio; two smaller projects show your fundamentals. Every one needs a question, checks and a useful conclusion.

THE PROJECT LADDER

Five levels. Real evidence.

Build from a clean spreadsheet to an end-to-end analysis. Each project answers a clear question and gives you something useful to discuss in an interview.

LV.1 · Week 4

Retail sales workbook

Excel · Power Query · PivotTables

Question: Which products and trading periods drive revenue in this synthetic retail dataset?

Public Online Retail II dataset (check licence); or clearly labelled synthetic data ↗

- Document raw/clean sheets, row counts and a data dictionary.
- Create revenue and units summaries with slicers.
- Check returns, missing customer IDs and duplicate rows explicitly.
- Ship .xlsx, a one-page PDF and three findings.

Interview proof: Can you clean and reconcile a spreadsheet without hiding the messy parts?

LV.2 · Week 8

Retail SQL investigation

PostgreSQL · SQL · GitHub

Question: Which categories perform best, and how does performance change over time?

Northwind sample (SQL Server scripts; adapt schema for PostgreSQL, or use the Contoso retail course files), or your own synthetic retail tables ↗

- Use customers, orders, order lines and products with known keys.
- Write 30 queries, including joins, CTEs and ranking windows.
- Include failed assumptions and checks that caught duplicate joins.
- Ship schema, loading instructions, queries and selected results.

Interview proof: Can you turn a question into a trustworthy query?

LV.3 · Week 13

Retail performance dashboard

Power BI · Star schema · DAX

Question: What changed this month, and which category deserves attention?

Reuse public or synthetic retail data and label its source clearly ↗

- Build a sales fact table with product and date dimensions.
- Validate revenue and comparison measures against SQL.
- Create two focused pages rather than ten cluttered ones.
- Ship PBIX + screenshots/PDF + 3-minute demo; public sharing is optional.

Interview proof: Can you model data and explain a measure, not merely draw a chart?

LV.4 · Week 19

NHS waiting-time trend brief

Python · pandas · Matplotlib

Question: How do selected published waiting-time measures vary over comparable periods?

NHS England RTT published statistics; inspect definitions and available periods ↗

- Select a small set of comparable published files and record metadata.
- Handle missing/suppressed values without treating them as zero.
- Generate clean data, summary tables and a readable trend chart.
- Explain that aggregate waiting-time data cannot establish causes or individual outcomes.

Interview proof: Can you make a public-data analysis reproducible and cautious?

LV.5 · Weeks 20–23

East Sussex insight capstone

SQL · Power BI · Excel checks · optional pandas

Question: How has one local indicator changed, and how does East Sussex compare with an appropriate benchmark?

East Sussex in Figures; use ONS as an alternative source ↗

- Choose a manageable indicator with stable geography and definitions.
- Document extraction, cleaning, joins, grain and reconciliation.
- Create a dashboard and one-page evidence-led briefing.
- Ship code, README, sources, limitations and a five-minute walkthrough.

Interview proof: Can you own a complete analysis and defend its limitations?

Portfolio honesty: project outcomes are not workplace achievements. Describe the report you built, the checks you performed and the findings you can verify. Only claim savings or workplace impact you actually measured.



ATTACK ON TITAN ARC

You don’t know the outcome. Apply anyway.

Use rejection as feedback, not a verdict. Keep your search matched to your skills, geography and eligibility—and keep improving the evidence you send.

ENDGAME

Your South-East job-search playbook.

Three search areas, clear boundaries and an application that matches the actual person specification.

01

South-East London & border

Set separate alerts for Bexley, Greenwich/Woolwich, Lewisham/Catford, Bromley and nearby Dartford. Check the actual office base. Oxleas and Lewisham & Greenwich are employer leads; reviewed local examples were experienced posts, not junior openings.

02

Brighton, Hove & Sussex

Look at reporting, MI, data-quality and people-data roles as well as Junior Data Analyst. EDF and L&G are historical entry-route examples; MetLife is a senior comparator. Monitor future roles, not expired adverts.

03

Wider London / nearby Kent

Canary Wharf, the City and Canterbury are separate search areas—not SE London vacancies. Treat each as a commute/attendance decision from your actual home, not a default promise of remote work.

Bookmark your search routes

NHS Jobs ↗ · Civil Service Jobs ↗ · Reed ↗ · Indeed ↗ · LinkedIn ↗

East Sussex County Council ↗

Alerts: **Junior Data Analyst · Reporting Analyst · MI Analyst · Information Analyst · BI Assistant · Data Quality Analyst.** Exclude Business Analyst roles unless the actual work is data analysis and interests you.

Starter target list · research leads, not confirmed openings

ORGANISATION	AREA	WHY INVESTIGATE
East Sussex Healthcare NHS Trust ↗	Eastbourne / Hastings	Information, performance and reporting teams
East Sussex County Council ↗	Lewes / county-wide	Performance, public-service data and reporting
University Hospitals Sussex NHS FT ↗	Sussex sites	Information and operational reporting; check exact base
Brighton & Hove City Council ↗	Brighton / Hove	Data, insight and performance roles
American Express ↗	Brighton / UK	Operations and analytics; some roles need prior experience
Legal & General ↗	Hove / UK	Reporting and financial-services analytics; check entry criteria

ORGANISATION	AREA	WHY INVESTIGATE
Bupa ↗	Brighton / UK	Healthcare/insurance reporting and insight
University of Sussex ↗	Falmer	Planning, student data and reporting

Check current vacancies, office base and essential criteria on each official site. This list does not assert that any employer is currently hiring juniors.

Application maths you control

Starting target: **5 well-matched applications + 2 genuine networking messages per week**, when suitable vacancies exist. Four weeks at that rate is 20 applications—not a predicted number of interviews.

After 15–20 matched submissions with no screening calls, review your CV, evidence, targeting and eligibility. If screens happen but tests fail, practise the failed skill. If interviews fail, review examples and communication.

Your tracker fields

Date · employer · role · URL · salary · location/office days · essential criteria · eligibility · CV version · project used · stage · follow-up date · result · feedback.

Read right-to-work and sponsorship conditions carefully. Your immigration status is not assumed in this plan. Never pay a recruiter a fee to secure an ordinary job.

A referral message you can make your own

```
Hi [Name], I’m building towards a junior data-analytics role, with a background in [your current work or study].

I’ve built [project + link] using [Excel / SQL / Power BI]. It answers [specific question], with [one verified finding or quality check].

I saw [role + reference] at [company]. If you feel my work is a fit, would you be open to a short conversation or pointing me towards the right application route?

Thanks for your time,
[Your name]
```

CV upgrades before your next application

- **Projects first:** question → tools → checks → actual result. Use numbers only when true.
- **Keep the stack visible:** Excel, SQL, Power BI; add Python when you can demonstrate it.
- **Show transferable skills honestly:** prioritisation, accuracy, teamwork and customer communication. Do not invent volumes, savings or analytical duties.
- **Use a clean one- or two-page UK CV:** no photo, skill bars or decorative columns. Tailor length to useful evidence.
- **Clickable links:** GitHub, flagship case study and demo. Check them while logged out.
- **Keep employment details accurate:** use truthful role titles, employment dates and responsibilities.

Optional NHS application pack

Only if targeting healthcare: read the NHS Data Dictionary ↗ and the methodology for your chosen published dataset. Learn what a numerator, denominator, reporting period, submission deadline and data-quality exception mean in that context. Do not claim access to clinical systems or that a public project replaces required NHS experience.

Optional finance / retail pack

Build a short glossary: revenue, gross margin, returns, stock turnover, customer/transaction grain and reporting period. For wealth/insurance roles, add the basic product context requested in that advert. Validate any formula before presenting it as a decision metric.

Apply earlier when appropriate: once you can independently clean, reconcile and explain an Excel workbook, consider genuine Excel-led entry roles whose essential criteria you meet. Broaden to SQL/BI roles as those projects become defensible. Week 13 is a planning milestone, not a ban on earlier applications.

South-East London employer watchlist

Use the official site's jobs/careers section. These organisations are leads to monitor, not verified current junior vacancies.

- Bexley Council · Bexley ↗
- Royal Borough of Greenwich · Greenwich / Woolwich ↗
- Lewisham Council · Lewisham ↗
- Bromley Council · Bromley ↗
- Oxleas NHS Foundation Trust · SE London / Dartford; check actual job base ↗
- Lewisham and Greenwich NHS Trust · SE London; check actual job base ↗

BOSS DIALOGUE

Interview question bank.

40 questions with answer guidance. Answer aloud before opening the guidance, and use your own examples rather than memorising a script.

SQL · 10 questions

▼ 1. What is the difference between WHERE and HAVING?

Answer direction: WHERE filters rows before aggregation; HAVING filters grouped results. Show a total-sales example.

▼ 2. What is the difference between INNER and LEFT JOIN?

Answer direction: Explain which rows are preserved, unmatched NULLs and the importance of the join key.

▼ 3. Why did revenue double after a join?

Answer direction: Check the grain and key uniqueness. A one-to-many join can duplicate fact rows; aggregate or fix the key, do not hide it with DISTINCT.

▼ 4. What is the difference between COUNT(*), COUNT(column) and COUNT(DISTINCT column)?

Answer direction: All rows, non-NULL values and distinct non-NULL values respectively.

▼ 5. How would you get the top three products per category?

Answer direction: Aggregate to product/category grain, then use ROW_NUMBER or DENSE_RANK depending on the tie policy.

▼ 6. How do you find customers with no orders?

Answer direction: LEFT JOIN + unmatched-key IS NULL, or NOT EXISTS; explain NULL risks with NOT IN.

▼ 7. When would you use a CTE?

Answer direction: To express logical steps clearly. It is not automatically a performance improvement.

▼ 8. How would you calculate month-on-month change?

Answer direction: Aggregate by month, use LAG and divide the difference by the prior value with zero handling. Consider absent months.

▼ 9. How do you handle missing values?

Answer direction: Decide what NULL means, use IS NULL and only COALESCE when the replacement is semantically justified.

▼ 10. How would you validate your result?

Answer direction: Check row counts, grain, key uniqueness, totals, sampled cases and an independent calculation.

Excel & data quality · 10 questions

▼ 1. When would you use XLOOKUP rather than a PivotTable?

Answer direction: A lookup returns related values by key; a pivot summarises data. Check key uniqueness and the desired grain.

▼ 2. How would you clean a messy CSV?

Answer direction: Preserve raw data, inspect types, dates, blanks and duplicates, then document and reconcile each transformation.

▼ 3. What is a repeatable Power Query workflow?

Answer direction: Typed import → cleaning steps → merge/unpivot → validation → refreshable output. Test on a replacement file.

▼ 4. What is the difference between a blank and zero?

Answer direction: A blank may be unknown or not applicable; zero is an observed value. They can change counts and averages differently.

▼ 5. How do SUMIFS and COUNTIFS help?

Answer direction: They aggregate by multiple criteria. Explain a date-bounded category-sales example.

▼ 6. How do you check a lookup is correct?

Answer direction: Use exact matching, inspect unmatched values, test duplicates and sample returned records.

▼ 7. How would you build a manager's weekly report?

Answer direction: Clarify decision, definitions and audience; automate import, reconcile totals, then present a small number of useful KPIs.

▼ 8. What is percentage-point change?

Answer direction: A rate moving from 10% to 12% rises by 2 percentage points or 20% relative—not 2% relative.

▼ 9. Why might Excel show dates incorrectly?

Answer direction: Locale, mixed text/date types or serial formatting; parse deliberately and inspect ambiguous dates.

▼ 10. What would you do if two reports disagree?

Answer direction: Compare sources, refresh times, grain, filters, date boundaries and definitions before deciding which is wrong.

Power BI, statistics & Python · 10 questions

▼ 1. Why use a star schema?

Answer direction: Clear fact/dimension grain, predictable relationships and reusable measures; explain your own model.

▼ 2. What is the difference between a measure and a calculated column?

Answer direction: A measure evaluates at query time in filter context; a column is computed per row and stored during refresh.

▼ 3. What does CALCULATE do?

Answer direction: Evaluates an expression in modified filter context. Demonstrate using a measure from your project.

▼ 4. How would you test a dashboard?

Answer direction: Reconcile KPIs, test slicers and relationships, inspect blanks, zero denominators, dates and accessibility.

▼ 5. When is median more useful than mean?

Answer direction: Often when a distribution is skewed or affected by outliers; explain both with a simple example.

▼ 6. Does correlation prove causation?

Answer direction: No. Confounding, selection bias and reverse causality may explain an association.

▼ 7. How do you explain uncertainty to a manager?

Answer direction: Describe the range, assumptions, sample/data limits and what additional evidence would change the decision.

▼ 8. How would you validate a pandas merge?

Answer direction: Inspect key uniqueness, use validate= where appropriate, track unmatched rows and reconcile totals and counts.

▼ 9. How do you make a notebook reproducible?

Answer direction: Record source versions and dependencies, use clear relative paths and re-run from a fresh environment in order.

▼ 10. Should you use Publish to web for a company report?

Answer direction: No for confidential data: it makes content publicly accessible. Use approved sharing, access controls and licensing.

Career change & behavioural · 10 questions

▼ 1. Why do you want to work in data analytics?

Answer direction: Use a truthful past → interest → concrete learning evidence → role-fit story. Avoid criticising retail work.

▼ 2. Tell me about an error you caught.

Answer direction: Use a real STAR example: what you checked, what you did and what actually happened. No invented savings.

▼ 3. How do you prioritise competing tasks?

Answer direction: Clarify urgency, impact, dependencies and stakeholder expectations. Use a genuine shift or project example.

▼ 4. Tell me about working with a difficult customer.

Answer direction: Show listening, clarity, boundaries, appropriate escalation and a factual outcome.

▼ 5. Describe a time you worked in a team.

Answer direction: Be specific about your contribution and recognise others; do not claim the whole team’s result as yours.

▼ 6. How do you explain technical work to non-technical people?

Answer direction: Start with the decision and meaning, use one clear visual, define units and check understanding.

▼ 7. What if a manager requests a misleading chart?

Answer direction: Explain the risk respectfully, show an honest alternative and document or escalate if needed.

▼ 8. What would you do with sensitive customer data?

Answer direction: Use only authorised access and purpose, minimise exposure, follow policy and ask the appropriate data owner if uncertain.

▼ 9. What is a weakness you are improving?

Answer direction: Choose a genuine manageable gap and give a concrete practice plan with evidence of progress.

▼ 10. What would you ask us?

Answer direction: Ask about the team’s data quality problems, first-90-day expectations, code/report review and support for junior analysts.

YOUR QUICK-REFERENCE DESK

Shortcuts & cheat sheets.

Keyboard shortcuts, syntax and practical reminders for every stage of the roadmap. Look up what you need, then put it into practice.

ALL-IN-ONE · OFFLINE PDF

Your analytics desk reference.

Eight compact chapters: Excel & Power Query, SQL, Power BI, statistics & quality, Python/pandas, Git, projects and interview preparation.

PDFs from trusted learning sources

Open a publisher's PDF in a new tab. Some browsers download the file instead. These links need internet; save a copy for offline use. Access checked 19 October 2026.

CustomGuide PDF · 7 pages

Excel

Keyboard shortcuts, formulas, tables and workbook skills. Windows-first; Mac and web keys differ.

Open PDF ↗
https://www.customguide.com/cheat-sheet/excel-cheat-sheet.pdf

SQL Tutorial PDF · 3 pages

SQL essentials

SELECT, joins, grouping and table operations. Mixed dialects: verify LIMIT/TOP and MINUS/EXCEPT before using.

Open PDF ↗
https://www.sqltutorial.org/wp-content/uploads/2016/04/SQL-cheat-sheet.pdf

LearnSQL PDF · 2 pages

SQL window functions

PARTITION BY, ranking, LAG/LEAD and window frames. Support differs by database.

Open PDF ↗
https://learnsql.com/blog/sql-window-functions-cheat-sheet/Window_Functions_Cheat_Sheet.pdf

Dataquest PDF · 16 pages

Power BI & DAX

DAX, Power Query, modelling, visuals, refresh and security. Use as a lookup, not an exam checklist.

Open PDF ↗
https://www.dataquest.io/wp-content/uploads/2025/01/Power-BI-Cheat-Sheet.pdf

Eric Matthes

PDF · 2 pages

Python fundamentals

Variables, lists, dictionaries, loops and functions. GitHub may download the PDF instead of displaying it.

Open PDF ↗
https://github.com/ehmatthes/pcc_3e/releases/download/v1.0.0/beginners_python_cheat_sheet_pcc.pdf

pandas project

PDF · 3 pages

pandas

Filtering, missing values, grouping, reshaping and joins. Check examples against your installed version.

Open PDF ↗
https://pandas.pydata.org/Pandas_Cheat_Sheet.pdf

GitHub Education

PDF · 2 pages

Git

Status, staging, commits, branches and remotes. Avoid destructive commands unless you understand their effect.

Open PDF ↗
https://education.github.com/git-cheat-sheet-education.pdf

Quick lookup

Open a topic for keyboard keys, useful code patterns and a short workflow checklist. Keyboard shortcuts depend on your platform and the active editor.

▼ Excel & Power Query

Make everyday spreadsheet work faster.

SHORTCUT / REMINDER	USE IT FOR
Ctrl + T	Turn a range into an Excel Table.
Ctrl + Shift + L	Toggle filters.
Ctrl + Arrow / Ctrl + Shift + Arrow	Jump to / select to the edge of a data region.
Ctrl + 1 / Ctrl + Alt + V	Format Cells / Paste Special.
F2 / F4	Edit a cell / cycle reference anchors while editing a formula.
Alt + = / Ctrl + `	AutoSum / show formulas.
Ctrl + ; / Ctrl + Alt + F5	Insert today's date / refresh all connections.
Alt, N, V	Open PivotTable options; press keys in sequence.

```
=SUMIFS(Sales[Revenue],Sales[Category],A2)
=COUNTIFS(Sales[Status],"Returned")
=XLOOKUP(A2,Products[ID],Products[Category],"Not found")
=IFERROR(B2/C2,"")
=TRIM(A2)
=MEDIAN(Sales[Revenue])
```

- Windows desktop shortcuts; laptop function keys may need Fn. Mac and web versions differ. XLOOKUP needs a supported Excel version.
- Power Query routine: connect → preserve raw data → set types and UK date locale → trim/clean → investigate blanks and duplicates → merge/append → reconcile → load → refresh.
- Merge adds columns by matching keys; append stacks rows. Check key uniqueness before a merge. Keep exception rows instead of silently deleting them.
- Formula examples assume the named tables and columns exist. IFERROR is not a substitute for checking the denominator.

Excel keyboard shortcuts ↗

▼ SQL & the T-SQL bridge

Remember the pattern, then check the result.

SHORTCUT / REMINDER	USE IT FOR
SELECT / WHERE	Choose columns / filter rows before grouping.
GROUP BY / HAVING	Aggregate by a defined grain / filter groups.
LEFT JOIN	Keep every left-hand row; watch for one-to-many multiplication.
COUNT(*) / COUNT(column)	Count rows / count non-NULL values.
IS NULL / COALESCE	Test missing values / choose a deliberate replacement.
WITH name AS (...)	Name a CTE to make query stages readable.
ROW_NUMBER / RANK / DENSE_RANK	Unique sequence / ties with gaps / ties without gaps.
LAG / LEAD	Read a previous / next row within an ordered partition.

```
WITH monthly AS (
  SELECT category, SUM(revenue) AS sales
  FROM sales
  WHERE sale_date >= '2026-01-01'
    AND sale_date < '2026-02-01'
  GROUP BY category
)
SELECT category, sales,
  DENSE_RANK() OVER (ORDER BY sales DESC) AS sales_rank
FROM monthly;
```

- Logical order, simplified: FROM/JOIN → WHERE → GROUP BY → HAVING → window calculations → SELECT → DISTINCT → ORDER BY → row limit. Write a CTE when you need to filter a window result.
- T-SQL uses TOP (n) rather than PostgreSQL LIMIT n. Date functions, casts and identifier quoting vary; portable concepts do not make every query portable.
- Use an explicit ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW frame for a row-by-row running total; order by a stable tie-breaker.
- Before trusting a join: check key uniqueness, unmatched rows, row count and totals. Never use DISTINCT simply to conceal a broken join. Practise UPDATE/DELETE only in a safe copy.

SQL reference ↗ · T-SQL bridge ↗

▼ Power BI, DAX & report operations

Fast authoring is useful. Reliable measures matter more.

SHORTCUT / REMINDER	USE IT FOR
Shift + ?	Show available Power BI Desktop keyboard shortcuts.
Ctrl + F6 / Tab / Shift + Tab	Move between sections / forward / backward.
Alt	Show ribbon keytips.
Ctrl + Enter	Submit a DAX formula.
Ctrl + G	Go to a line in the DAX formula bar.
Alt + Up / Down	Move a line in the DAX formula bar.
Ctrl + G (canvas editing)	Group selected visuals; the same keys have a different formula-bar meaning.
Esc	Exit a visual or layer.

```
Revenue = SUM(Sales[Revenue])
Orders = DISTINCTCOUNT(Sales[OrderID])
Average Order Value = DIVIDE([Revenue], [Orders])
Revenue All Categories =
    CALCULATE([Revenue], REMOVEFILTERS(Products[Category]))
```

- Create these as measures; tables and columns are illustrative. A measure responds to filter context. A calculated column is evaluated row by row at refresh and stored.
- Model first: define fact-table grain, use dimension keys, prefer one-to-many relationships and deliberate filter direction. Confirm the date table before time-intelligence work.
- Release check: reconcile KPI totals → test slicers and blanks → add clear units/alt text → inspect mobile view → run Performance Analyzer → test RLS roles.
- Service checklist: workspace access, gateway where needed, credentials, scheduled refresh, refresh-failure ownership and sharing permissions. Desktop success is not proof of Service refresh or security.

Official Power BI keyboard shortcuts ↗ · Official DAX reference ↗

▼ Statistics, metrics & data quality

Keep the denominator, uncertainty and business question visible.

SHORTCUT / REMINDER	USE IT FOR
Mean / median	Mean is sensitive to outliers; median is useful for skewed values.
Weighted mean	SUM(value × weight) / SUM(weight); validate the weights.
Percentage change	(new − old) / old; format as a percentage and define what happens at old = 0.
Percentage points	30% − 20% = 10 percentage points; the relative increase is 50%.
Conversion rate	Conversions / eligible opportunities over a defined period.
IQR	Q3 − Q1; inspect the distribution before using outlier rules.
Sample / population spread	Use STDEV.S for sample estimation; STDEV.P for the full population described.
Correlation	Association is not evidence of causation.

```
BEFORE ANALYSIS
Question → audience → metric → grain → period

AFTER EVERY TRANSFORMATION
Rows → keys → missingness → totals → exceptions

BEFORE SHARING
Source → assumptions → limits → privacy → owner
```

- A p-value is not the probability that the null hypothesis is true. Explain the test assumptions, effect size and uncertainty instead of treating 0.05 as an automatic business decision.
- A frequentist 95% confidence procedure covers the fixed true parameter in 95% of repeated samples under its assumptions; it is not a guarantee about an individual outcome.
- Missing is not automatically zero. Investigate causes and document any exclusion or imputation. Reconcile with an independent method where possible.
- Use only permitted public/synthetic practice data. Minimise personal information, keep raw data unchanged and document transformations and retention.

UK data-protection principles ↗

▼ Python, pandas & plotting

Inspect first. Transform deliberately. Make the notebook reproducible.

SHORTCUT / REMINDER	USE IT FOR
df.head() / df.info()	Preview rows / inspect types and missing counts.
df.isna().sum() / df.duplicated().sum()	Count missing values / exact duplicate rows.
df.loc[mask, columns]	Select explicit rows and columns; avoid chained assignment.
pd.to_datetime(..., errors="coerce")	Parse dates; inspect new NaT values instead of ignoring them.
groupby(...).agg(...)	Summarise at a defined grain.
merge(..., validate="many_to_one")	Enforce the expected join relationship.
pd.concat([...], ignore_index=True)	Append matching tables.
sort_values(...) / to_csv(..., index=False)	Order the result / export without the index.

```
import pandas as pd
import matplotlib.pyplot as plt

sales = pd.read_csv("sales.csv")
products = pd.read_csv("products.csv")
out = sales.merge(products, on="product_id",
    how="left", validate="many_to_one", indicator=True)
assert len(out) == len(sales)
print(out["_merge"].value_counts())
summary = out.groupby("category", dropna=False)["revenue"].sum()
summary.sort_values().plot.barh()
plt.xlabel("Revenue (£)")
plt.tight_layout()
plt.savefig("revenue_by_category.png", dpi=150)
```

- Example assumes the listed CSV columns exist and revenue is numeric. Resolve unmatched keys before interpreting category totals. Decide how missing revenue should be handled before summing.
- Create a virtual environment; install required libraries and record versions. Restart the kernel and run every cell in order before sharing.
- Python basics to recall: = assigns; == compares; indentation defines blocks; lists are ordered; dictionaries map keys to values. Use functions for repeated transformations.
- Name chart units, period and source. Use bars for category comparisons, lines for time series and scatter plots for relationships.

pandas API ↗ · Matplotlib cheat sheets ↗

▼ Setup, Git & README essentials

Keep the daily loop small and your public repository safe.

SHORTCUT / REMINDER	USE IT FOR
git status / git diff	Inspect changed files / inspect unstaged changes.
git add file / git diff --staged	Stage a specific file / inspect what will be committed.
git commit -m "Message"	Save a descriptive local checkpoint.
git pull --rebase	Update from the tracked remote; only use when you understand local history and conflicts.
git push / git log --oneline	Publish commits / review compact history.
git restore --staged file	Unstage a file without discarding the working copy.
.gitignore	Exclude credentials, private data, environments and generated files.
Markdown: # / ## / - / [title](URL)	Title / subheading / list / descriptive link.

```
git status
git diff
git add analysis.sql README.md
git diff --staged
git commit -m "Add validated monthly sales analysis"
git pull --rebase
git push

# README headings
# Question | Data | Method | Checks
# Findings | Limits | Reproduce | Sources
```

- Set up folders for raw data, cleaned data, SQL, notebooks, reports and screenshots. Power BI Desktop needs Windows access. Check Excel/Power Query version support before starting.
- Do not use reset --hard, force push or history rewriting as a routine shortcut. Inspect files before staging; do not commit secrets or confidential data.
- If a secret is exposed, revoke/rotate it immediately; deleting the latest file does not remove earlier Git history.
- Add install/run steps, data source and licence, expected outputs and a screenshot. Check repository and demo links while logged out.

GitHub Hello World ↗ · Markdown reference ↗

▼ Projects, capstone & handover

A repeatable workflow for every portfolio project.

SHORTCUT / REMINDER	USE IT FOR
1 · Define	Write one useful question, its reader, the metric and success criteria.
2 · Source	Record source, licence, date, grain and limitations.
3 · Profile	Check types, ranges, blanks, duplicate keys and suspicious values.
4 · Transform	Keep raw inputs unchanged; document and reproduce each step.
5 · Validate	Reconcile counts, joins and key totals independently.
6 · Explain	Show three findings, an actionable recommendation and limits.
7 · Deliver	Package the workbook/query/report/notebook, README and screenshots.
8 · Handover	Explain refresh, access, definitions, exceptions and who owns failures.

```
CASE STUDY IN SIX LINES
Question: What decision does this support?
Data: Where did it come from and what does one row mean?
Method: What did I change or calculate?
Checks: How do I know the result is sound?
Finding: What changed, by how much, over which period?
Limit: What can this analysis not establish?
```

- Portfolio ladder: retail Excel → retail SQL → retail Power BI → public NHS pandas analysis → end-to-end regional capstone. Lead with your strongest three pieces.
- Public healthcare projects need dataset definitions, reporting-period rules and published methodology. They do not prove experience with internal clinical systems.
- Use screenshots, a PDF and a short demo if live sharing is unsuitable. Never use Publish to web for confidential or personal data.
- Label synthetic work honestly. Describe what you built and verified; do not claim fictional savings, workplace deployment or business impact.

Roadmap projects ↗

▼ Revision, interviews & applications

Short preparation routines—not shortcuts around understanding.

SHORTCUT / REMINDER	USE IT FOR
20-minute recall loop	5 minutes explain → 10 minutes solve → 5 minutes check and log one gap.
60-second introduction	Background → why analytics → evidence → role fit.
Project answer	Question → data → method → validation → finding → limitation.
Behavioural answer	Situation → task → action → result; describe your own contribution.
SQL test routine	Confirm grain and keys → write the query → test NULLs/ties → explain checks.
Role screening	Verify location, level, essential skills, deadline and genuine paid employment.
CV evidence	Tool + question answered + checks + truthful result; link the project.
Application tracking	Role, employer, URL, closing date, fit, tailored evidence, status and follow-up.

```
WEEKLY REVIEW
1. What can I now do without copying?
2. Which result have I independently checked?
3. What is the smallest remaining blocker?
4. Which role criteria can I demonstrate today?
5. What is my next repeatable practice session?
```

- Apply when you meet a role's essential criteria; the roadmap week is not a ban on earlier applications. Excel-led reporting roles can have different requirements from SQL/BI roles.
- Keep one main resource per tool. After a lesson, close it and reproduce the task on a different small dataset. Extend the calendar when needed rather than abandoning the habit.
- Use truthful dates, responsibilities and results. A personal project is not employment experience and a certificate is not proof of independent delivery.
- During an interview, clarify assumptions, explain uncertainty and say how you would verify an unfamiliar detail. Do not bluff or use unauthorised assistance.

UK data analyst career guidance ↗

Use a shortcut to save a click—not skip a check. Validate keys, row counts, units and totals before trusting a result. A command you can explain is more useful than a list you have memorised.

PLAN STRESS-TEST

Seven ways this plan could fail—and the guardrails.

Protect your time, avoid tutorial overload and build work you can trust.

01 · Too many tools at once.

Sequence the stack. Excel/SQL/Power BI lead; Python follows. Leave unrelated tools and advanced specialisms until your target roles justify them.

02 · A 28-week deadline ignores shift work.

The budget is 336 hours. Use the minimum-week plan and extend the calendar after heavy rotas or illness.

03 · Tutorial completion can hide weak understanding.

Every arc has a closed-book or reproducibility boss. Roughly 70% of practice time is spent doing and checking.

04 · A beautiful report can be numerically wrong.

Reconcile row counts, grain, joins and KPI totals with an independent method in every project.

05 · Certification and public sharing can create cost barriers.

PL-300 is optional. Screenshots, PDF exports, PBIX files and a demo can be enough to show evidence without a paid sharing plan.

06 · Local-only search may be too narrow.

Widen geography and role titles where the commute and criteria work. Employer names are leads, not claims of live vacancies.

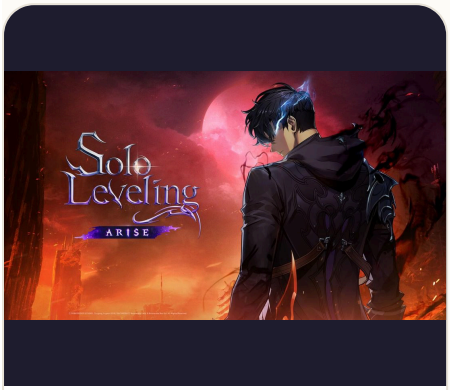
07 · Experience can be oversold.

Describe your experience honestly and label public or synthetic projects clearly. Use numbers only when you can explain how they were measured.

WHY YOU'LL FINISH

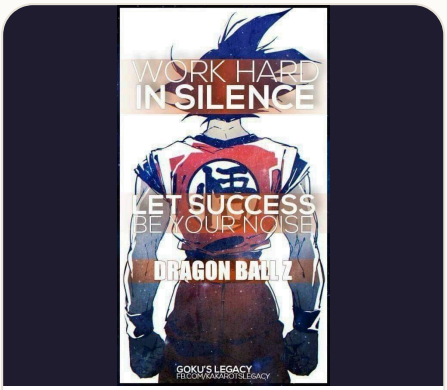
Keep showing up.

You don't need to master everything today. Finish one useful task, understand it, and return for the next.



E-RANK → S-RANK

A result you can explain is a level you keep.



WORK HARD IN SILENCE

A calm 40-minute build beats another saved playlist.



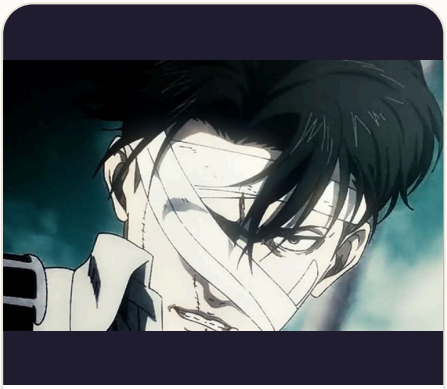
ROCK LEE · FUNDAMENTALS

You do not need to feel gifted to practise carefully.



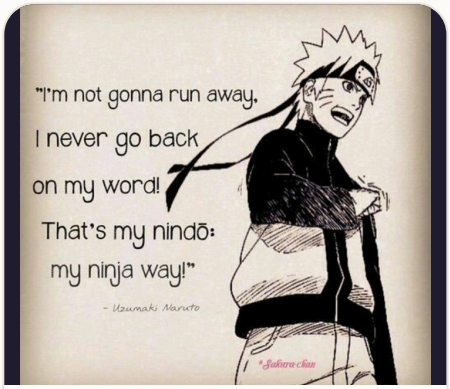
PLUS ULTRA

Go beyond the chart: validate the numbers underneath.



APPLY ANYWAY

Send the matched application. Review the feedback.



YOUR NINDŌ

Keep the promise small enough to repeat.

Your first role is not your final form.

The goal is not to know every tool. It is to become someone who can ask a useful question, check the data and explain the answer.

DUE DILIGENCE

Sources & further reading.

Evidence where it matters; clear boundaries where it is a recommendation.

1. Prospects · Data analyst role, skills and salary ↗ — UK career guidance; salary figures are national indicators, not local guarantees.
2. Excel for Data Analytics · Luke Barousse ↗ — Main Excel course; select chapters for the skill you are practising.
3. SQL for Data Analytics · Luke Barousse ↗ — Analyst-focused SQL foundations and project.
4. Intermediate SQL · Luke Barousse ↗ — Selected window-function, date and cleaning lessons.
5. Power BI for Data Analytics · Luke Barousse ↗ — Data preparation, modelling, DAX and dashboard practice.
6. Python for Data Analytics · Luke Barousse ↗ — Python and pandas practice; use current course notebooks.

Using this roadmap

Adapt the pace to your experience and commitments. This is a learning plan, not a guarantee of employment. Check current role requirements, salaries and application deadlines before applying.

Save the HTML to use the roadmap offline. External videos and documentation require internet access. Progress is stored in your browser; export a backup before switching devices. The PDF is a static reading copy, and Excel progress is tracked separately.

Anime artwork belongs to its respective rights holders. Reuse is subject to those rights.

Additional references

Use official documentation for current features, access controls and exam requirements. Course interfaces and files can change; use the latest materials from the course provider.

- Excel course outline ↗ — Excel lesson files and chapter list. Use the repository ZIP or alternate file link if an individual workbook fails to open.
- SQL foundations ↗ — PostgreSQL setup, free starter files and project resources. Course datasets are historical, not current hiring evidence.
- Intermediate SQL ↗ — Window-function and analysis practice, with a fictional retail dataset.
- Power BI course outline ↗ — Power Query, DAX and project files. Use the provider’s complete download link for large files.
- Python course outline ↗ — Current notebooks for pandas, cleaning and visualisation.
- Current PL-300 study guide ↗ — The official exam objectives. Review them separately if you choose certification.
- T-SQL learning path ↗ — SQL Server querying fundamentals for the short dialect bridge.
- T-SQL window lesson ↗ — OVER, ranking, aggregate and offset functions.
- UK GDPR principles ↗ — Official guidance on purpose, minimisation, accuracy, security and accountability.

Operation Data Analyst · UK South East

Ask a useful question. Check the data. Explain the answer.

142 checkpoints · 9 seasons · 5 projects · 336 planned hours

0% complete · E-Rank

October 2026 · Save the HTML for offline use · Keep a progress backup.